2.2 Interim Digital Infrastructure (While Building Is Under Construction)

LMS Server	Dedicated server for Moodle — 16-core, 32GB RAM, 2TB SSD. Requisition through CE/IT within minor works budget.
Video Recording Studio	2 cameras + ring light + microphone + green screen — INR 1-2 L. Serving and retired engineers record subject lectures for Moodle repository.
Smart Classroom	Existing PRTI room retrofitted: projector, webcam, hybrid delivery capability. INR 1-2 L per room.
Computer Lab (Interim)	20 decommissioned APTRANSCO IT nodes — AutoCAD, PSCAD, SAP hands-on training. Near-zero cost.

3.2 Moodle Mobile App — Full Feature Set

App Feature	APTRANSCO Use Case	Priority
Offline Mode	Field engineers at remote substations download modules; sync on return to connectivity	Critical — field staff
Push Notifications	Deadline alerts, new module releases, assessment reminders	High
Video Streaming	NPTEL IIT lectures and OEM training videos (ABB/Siemens/Schneider)	High
Assessments / Quizzes	Pre/post training tests — auto-tracked to Moodle gradebook	High
Progress Dashboard	Employee: own hours. Manager: team completion % across zone	High
Calendar Integration	Training schedule auto-syncs to employee device calendar	Medium
Discussion Forums	Cross-zone technical forums — community of practice	Medium

3.3 SSO Integration — Options by Existing System

Method	When to Use	Complexity
LDAP / Active Directory	APTRANSCO uses Windows AD for employee login — most common in PSUs	Low — 2 days
SAML 2.0	Using Keycloak	Medium — 1 week

Method	When to Use	Complexity
SAP HR API sync	Employee master data in SAP — auto-provision Moodle from SAP HR	Medium — 2 weeks

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SAP HR API sync	Employee master data in SAP — auto-provision Moodle from SAP HR	Medium — 2 weeks

3.4 Deployment Specifications

Go-Live Target	30 days from internship onboarding Pilot: 100 AEs on Q2 technical track.
Server Specification	16-core, 32GB RAM, 2TB SSD. Handles 3,000+ concurrent users. Hosted on APTRANSCO internal server.
Intern Scope	Deploy Moodle, configure SSO, build course categories, import NPTEL SCORM packages, create competency tracking dashboards, build RoTI reporting module.
Content from Day 1	All Zoom recordings + existing PRTI materials uploaded as archived courses. Immediate value before new content is created.
3-Year Cost Estimate	Server INR 3-4 L one-time.

4. Content Strategy — ISGF, NPTEL, CBIP, OEM, TLA & Knowledge Repository

4.1 India Smart Grid Forum (ISGF)

Type	Public-Private Partnership under Ministry of Power — not commercial. GoI-backed.
Why APTRANSCO	Transmission-specific: Smart Grid, SCADA, Cybersecurity, Renewable Grid Integration, EV, Grid-Scale Storage — directly aligned with AP grid modernisation.
Certification	ISGF certificates recognized for career progression and regulatory compliance.
Cost	Near-zero for utility members. One-time institutional membership registration.
Tracking	Employer dashboard: employee enrolments, completions, certification status.
Integration	ISGF course links embedded in Moodle learning paths.
Action	CE/PRTI to email: info@isgf.in — 'APTRANSCO Institutional Membership'. JMD-level letter to ISGF Chairman carries stronger weight.

4.2 NPTEL Swayam — IIT-Branded Certification (Primary Platform)

Joint initiative of all IITs and IISc, funded by MHRD. Highest-quality free technical education in India. IIT brand at near-zero cost per employee.

Course Track	Target Cadre	Cert Cost
Power Systems Analysis	AEs, DEs, SEs	**INR 1,500**
SCADA & Industrial Automation	SCADA, MRT, EMS staff	**INR 1,500**
High Voltage Engineering	EHT, line maintenance staff	**INR 1,500**
Renewable Energy Integration	Planning, SLDC engineers	**INR 1,500**
Project Management for Engineers	SEs, EEs in project roles	**INR 1,500**
Data Analytics & ML Basics	IT, planning, analytics	**INR 1,500**
Cybersecurity Fundamentals	All staff with system access	**INR 1,500**
Leadership & Org Management	SE and above	**INR 1,500**
Electrical Safety & Standards	All technical staff	**FREE (audit)**
Financial Management — Utility	Accounts, Finance officers	**INR 1,500**

NPTEL-Moodle Integration: NPTEL courses export as SCORM 1.2 — embed directly into APTRANSCO Moodle. Employee starts in Moodle; IIT delivers content; completion auto-syncs back.

500 employees x 2 certifications = INR 15-20 Lakhs — IIT-branded certification for the cost of a single NPTI external batch.

Full APTRANSCO control	Owns data, workflows, content. No vendor lock-in. No change request charges.
Intern-buildable	NRI Institute / NIRF-ranked CSE interns deploy, configure, maintain within 30 days.
NPTEL / SCORM compatible	NPTEL SCORM 1.2 packages embed directly. Employee learns on APTRANSCO LMS; IIT provides content.
ISGF / CBIP linkable	External course links hosted as resources in Moodle learning paths.